

Catherine Azelby

[Portfolio](#) | [Email](#) | [LinkedIn](#) | (+1) 609-605-7772 | Willing to Relocate

Education

Northeastern University

Sept 2022 - Dec 2026

Candidate for a Bachelor of Science in Computer Science and Media Arts (Animation)

GPA: 3.730, Dean's List

Technical Skills

Animation

Maya, Unreal Engine, Blender, Houdini Solaris, Adobe Substance Painter, XGen, Marvelous Designer

Programming & Tech

Python, OpenUSD, MaterialX, Unity, C#, UE Blueprints, C++, MEL, Git, Gitlab, Perforce, AWS

Art & Other

Adobe Photoshop, Adobe Illustrator, Adobe After Effects, Adobe Premiere Pro, Jira, Confluence, MS Office

Work Experience

Lucasfilm Industrial Light & Magic (ILM)

Jun 2026 - Aug 2026

Advanced Development Group: Virtual Production Content Pipeline Intern

San Francisco, CA

- Developed a custom MaterialX node graph in Houdini Solaris to implement a proprietary ILM shading model to support previs and postvis workflows.
 - Ran LookDev tests and documentation across DCC renderers including USDView Storm, Houdini Karma, Unreal Engine, and RenderMan.
 - Created custom tools to assist material authoring and editing within Houdini such as a MaterialX UDIM Editor tool.
- Automated shader translation pipelines, developing custom graphs and workflows to convert glTF assets to ILM-compliant USD MaterialX assets.

MIT Department of Mechanical Engineering: LEAP Group

Jan 2026 - May 2026

Game Developer & Research Assistant

Boston, MA

- Worked in Unity to expand a custom desktop authoring platform seamlessly linked to a student-facing learning application. Features included:
 - AWS S3 data persistence mechanics to securely handle user files and private image uploads.
 - Dynamic 3D measurement generator, context-aware isolation tools, and custom box-collider scaling handles.
 - Instructor portal featuring secure authentication, direct cloud publishing flows, and student access-key validation.
- Created optimized, high-fidelity game assets 3D modeled in Blender and textured with Substance Painter.

HyLite Studios

Jun 2025 - Oct 2025

3D Visual Designer

Remote

- 3D modeling and texturing platform-compliant assets in Blender for Meta Horizon Worlds. Ensuring optimization for real-time performance while maintaining visual appeal.

Northeastern University

Jun 2025 - Oct 2025

Comic Illustrator

Remote

- Managing all aspects of art and illustration for "Chloe and the Locked Library" utilizing Photoshop, including character design, paneling, and cover illustration, ensuring timely delivery and alignment with the writer's vision.

Northeastern University XR Ed Lab

Jan 2025 - Jun 2025

Virtual Reality Developer & Research Assistant

Boston, MA

- Designed the prototype for a virtual reality field trip with Unreal Engine 5 and Meta Quest Developer Platform. Created an interactive car-crash simulator meant to engage and teach high school students about the physics concept of impulse.
- Collaborated with a local high school to conduct focus groups and interviews with teachers evaluating the effectiveness of VR learning experiences. Findings were synthesized and contributed to a paper submitted and accepted into the Meaningful XR Conference.

Projects

Chasm's Call: Challenge Entry: Utilized a unique workflow combining Blender character animation with an Unreal Engine environment and compositing in After Effects.

CameraLink: Maya/Unreal Plugin: Enables easy camera transfer between Autodesk Maya and Unreal Engine using industry-standard USD composition. Created with Python and C++.

Leadership, Involvement, and Honors

ASWF Dev Days: MaterialX Project Contributor

Sep 2026

- Contributed to the open-source MaterialX project by enforcing MaterialX specification rules for uniform node input validation in the Graph Editor

ACM SIGGRAPH Vancouver 2025 Student Volunteer

Aug 2025

- Assisted in providing conference support, guidance, and organization to guests attending the SIGGRAPH conference.

24 Hour Animation Challenge Top 50 Finalist

Oct 2025

- Worked with a team of 5 to create a 30-second animated film within 24 hours. The final film placed 29th out of over 700 teams.

MIT Reality Hack: Best Lifestyle World with Meta Horizon Worlds

Jan 2025

- 3D artist, modeler, and presentation lead for the winning team of the Meta Horizon Worlds track at MIT Reality Hack 2025.

Games, Art, and Play: Japan Study Abroad

July 2024

- Study abroad focusing on game design applications and principles. Created a visual novel dress-up game in Python and Photoshop.

Northeastern Animation Club President

April 2024 - April 2026

- Oversees club events, outreach, and activities. Plans events such as speaker events, film festivals, and alumni networking.

Northeastern College of Arts, Media, & Design Student Ambassador

Sep 2025 - April 2026

- Assisted with undergraduate recruitment and marketing efforts by sharing personal experience and advice.